```
void loop() {
  int hue = map(temperature(), hot, cold, hotColor, coldColor);
  hue = constrain(hue, hotColor, coldColor);
  fadeToHSB(blinkM, hue, 255, brightness);
  delay(10000);
}

float temperature() {
  float voltage = (analogRead(temperaturePin) / 1024.0) * 5.0;
  float celsius = (voltage - 0.5) * 100.0;
  float fahrenheit = (celsius * 1.8) + 32.0;
  if (degreesF == true) return fahrenheit;
  else return celsius;
}

void stopScript(byte address) {
  Wire.beginTransmission(address);
  Wire.write('o');
  Wire.endTransmission();
}

void setFadeSpeed(byte address, byte fadespeed) {
  Wire.beginTransmission(address);
  Wire.write('f');
  Wire.write(fadespeed);
  Wire.endTransmission();
}

void fadeToHSB(byte address, byte hue, byte saturation, byte brightness) {
  Wire.beginTransmission(address);
  Wire.write('h');
  Wire.write(hue);
  Wire.write(saturation);
  Wire.write(brightness);
  Wire.endTransmission();
}
```

Source Code Summary

The first line of our sketch is necessary for using the Wire library that will allow the Arduino and BlinkM to talk to each other using a protocol called I2C, discussed at length in Chapter 10. Since we don't need to worry about that for now, let's jump to the variable declarations and see what they do.

```
const int blinkM = 0x09;
const int temperaturePin = A0;
```

These first two lines set up the locations for the BlinkM and our temperature sensor. The first is an address to identify which BlinkM to talk to. This value comes as the default and if we only use one BlinkM, will not need to be changed. The second line establishes that our temperature sensor is connected to pin A0.